

Maharashtra State Board of Secondary and Higher Secondary Education

General Science — Standard Seven (Grade 7)

Chapter 6: Measurement and Physical Properties of Substances

Many substances around us exist as mixtures of two or more components. In chemical laboratories and industries, it is often necessary to separate the pure constituents of a mixture. One of the most effective and simple methods used for this purpose is **Chromatography**.

Let us try this: Separation of components of ink by Paper Chromatography

Apparatus & Materials: A 250 ml glass beaker, a strip of Whatman filter paper (or blotting paper), blue writing ink, a pencil, a ruler, and water (or chalkstick as an alternative).

Procedure:

1. Take a rectangular strip of filter paper measuring approximately 15 cm in length and 2 cm in width.
2. Using a pencil, draw a light horizontal line across the strip about 2 cm above the bottom edge.
3. Place a small concentrated drop of blue ink in the center of this line using a fine dropper or nib and let it dry.
4. Pour a small quantity of water into the beaker so that it reaches a depth of about 1 cm.
5. Suspend the paper strip vertically inside the beaker with the help of a split cork or clip, so that the bottom edge dips into the water, but the ink spot remains strictly above the water level.
6. Observe the apparatus quietly as the water begins to rise up the paper strip through capillary action.
7. As the water solvent moves upward past the ink spot, students will observe that the original blue spot begins to separate into distinct colored bands (e.g. violet, cyan, or pink) at different heights on the strip.

Scientific Principle & Learning Connection:

The different dye components of the ink have different solubilities in the solvent (water) and different degrees of adsorption to the filter paper fibers. The component which is more soluble ascends faster and reaches a greater height, while the less soluble component lags behind. This concrete experiment proves directly that the blue ink is not a single pure substance, but a mixture of multiple distinct color pigments.

Chapter 6: Measurement and Physical Properties (Continued)

Classroom Teaching Suggestions for Teachers:

- Teacher should conduct a discussion in the classroom about different mixtures seen in daily life, such as tea, sherbet, and air.
- Use a diagram from page 45 to show the arrangement of particles in solids, liquids, and gases.
- Teacher should show students a chart depicting various lab instruments.
- Ask students to form small groups and discuss why pure water is a compound while salt water is a mixture.
- Teachers should do an experiment demonstrating melting of ice when convenient.

Exercises and Questions:

1. What is meant by chromatography?
2. Which property forms the basis for separating components in chromatography?
3. Give two real-world examples of mixtures separated by physical methods.

Chapter 19: Properties of a Magnetic Field

A magnet attracts magnetic substances towards itself. The region around a magnet where its influence can be detected is called its magnetic field. Since magnetic forces cannot be seen with naked eyes, we use physical methods to trace the lines of magnetic force.

Activity: Tracing Magnetic Lines of Force using Iron Filings and Floating Pins

Materials: A strong bar magnet, a stiff sheet of white cardboard or glass plate, fine iron filings, a sifter, small steel pins, a plastic bottle or shallow water bowl.

Instructional Procedure:

1. Place a strong bar magnet horizontally in the center of a wooden table.
2. Place a sheet of smooth white cardboard directly over the bar magnet.
3. Using a fine sifter, sprinkle iron filings evenly and thinly over the entire surface of the cardboard.
4. Instruct students to gently tap the edge of the cardboard with a fingertip 3 to 4 times.
5. Alternatively, place small steel pins inserted through small styrofoam discs floating on water in a flat shallow dish, placed adjacent to the magnet.
6. Direct student observation: Notice how the scattered iron filings immediately twist and arrange themselves into distinct, continuous curved curves stretching from one pole of the magnet to the other.
7. Guide students to observe the concentration of lines at the North and South poles compared to the center of the magnet.

Learning Connection:

Each iron filing becomes a tiny induced magnetic dipole and aligns along the direction of the magnetic field vector at that point. The curved paths formed by the filings visualize Michael Faraday's lines of magnetic force. The high density of lines at the poles demonstrates that magnetic field intensity is strongest near the poles.

Chapter 19: Magnetic Field (Continued)

Observe and Discuss:

Look at the picture of the Earth's magnetic field on the board. Observe how compass needles point north.
Try this at home: Take a needle, stroke it with a magnet, and see if it can attract small paper clips.

Supplementary Teaching Resources:

- Educational Video CD: 'Mysteries of Magnetism' (DIKSHA portal QR Code).
- Activity Worksheet No. 7: Magnetism Crossword Puzzle.
- Laboratory apparatus: Magnetic compass, Horseshoe magnet, iron stand.

Chapter 2: Plants: Structure and Function

What helps us to easily identify the plants around us?

Observe the plants around you and discuss their parts with your classmates.

The root, stem, leaves, flowers, fruits, etc. of different plants are different. We can identify plants with the help of these different characteristics. Let us now examine these plant organs in more detail.

In the primary grades, we have learned that plants have various organs that perform specific vital functions. Each organ is adapted to the environmental conditions in which the plant grows.

Maharashtra State Board of Secondary and Higher Secondary Education

Mathematics — Standard Six (Grade 6)

Chapter 1: Basic Concepts in Geometry — Angle Bisector

A ray which divides an angle into two equal parts is called the **bisector** of that angle. We can construct an accurate angle bisector using a ruler and compasses.

Activity: Constructing an Angle Bisector using a Compass and Ruler

Materials & Instruments: Compass box containing ruler, pencil, compass with sharp pencil tip, and blank paper.

Step-by-step Construction Procedure:

1. Draw any angle $\angle ABC$ of measure 70° on the paper using a protractor and ruler.
2. Place the metallic point of the compass at vertex B . Taking any convenient radius, draw an arc to cut ray BA at point P and ray BC at point Q .
3. Now, place the point of the compass at point P . Taking a radius more than half the distance between P and Q , draw an arc inside the angle.
4. Without changing the radius, place the compass point at point Q and draw another arc intersecting the previous arc. Name the point of intersection as O .
5. Draw a straight ray from vertex B passing through point O using a ruler. Ray BO is the required angle bisector of $\angle ABC$.
6. Verification: Measure $\angle ABO$ and $\angle CBO$ using a protractor. Both angles measure exactly 35° , demonstrating that ray BO bisects $\angle ABC$ into two congruent angles.

Mathematical Principle & Learning Connection:

Points on the angle bisector are equidistant from the two arms of the angle. By maintaining equal radii from points P and Q , point O forms two congruent triangles ($\triangle BPO \cong \triangle BQO$ by SSS congruence), ensuring $\angle PBO = \angle QBO$.